



GAMING & INTERACTIVE ENTERTAINMENT

VIDEO GAME SALES & MARKET TRENDS

Data Visualization & Analytics | Capstone Project | SecE_G15

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GitHub: [Git Repo](#)

Tableau: [Live Dashboard Link](#)

Why Does This Problem Matter?

The business case for data-driven decision-making in gaming

SECTOR CONTEXT

The video game industry is capital-intensive with multi-year development cycles. Publishers routinely risk \$100M+ budgets with no data-driven framework to guide decisions.

Decision-makers: Publishing Executives, Marketing Directors, Portfolio Managers

CORE PROBLEM

Which genres, platforms & regions yield the highest ROI? Does critical acclaim actually drive commercial success — or is it a myth?

11,559 games analyzed · \$8,917.53M total sales · 1980–2016

OBJECTIVE

Replace gut-feeling with hard historical data. Build a framework that quantifies how genre, platform, region & review scores dictate commercial success.

Output: Interactive Tableau Dashboard + Actionable Recommendations

Source → Pipeline

Python ETL · VGChartz + Metacritic · notebooks/01_ingestion.ipynb · 02_cleaning.ipynb

16,719

Raw Rows

16

Original Columns

20

Final Columns (+ 4 engineered)

1980–2016

Time Period Covered

RAW DATA

16,719 rows
16 cols

CLEAN

Drop 2 rows
Impute 269 years
Remove outliers

TRANSFORM

Standardise text
Cast types
Deduplicate

ENGINEER

Era · Score_Gap
Sales_Region_Dom
Global_Calc

KEY COLUMNS (Data Dictionary)

- Score_Gap
- Era
- Sales_Region_Dominant
- Global_Sales_Calculated

Critic_Score – (User_Score × 10) → measures critic vs player divide

Decade grouping of Year_of_Release (1980s, 1990s, 2000s, 2010s)

Single region contributing highest % of sales per title

NA + EU + JP + Other → used as validation column

What Are We Measuring & Why?

Reference: notebooks/03_edu.ipynb · 04_statistical_analysis.ipynb

Total Global Sales (\$M)

```
df['Global_Sales'].sum()
```

WHY IT MATTERS

Establishes total addressable market and macro industry health.

Maps to: Viability of physical game releases

Score Gap

```
Critic_Score - (User_Score × 10)
```

WHY IT MATTERS

Identifies cult classics players love despite poor critical PR.

Maps to: True drivers of player retention over critical acclaim

Regional Market Share (%)

```
(NA_Sales / Global_Sales_Calc) × 100
```

WHY IT MATTERS

Dictates where to spend localised marketing budgets.

Maps to: Targeted regional launch strategies

Platform Hit Volume

```
df.groupby('Platform')['Global_Sales'].sum()
```

WHY IT MATTERS

Proves which hardware ecosystem has the largest active buying base.

Maps to: Risk mitigation by launching on dominant consoles

What the Data Actually Says

Reference: [notebooks/03_edu.ipynb](#)

01 NA Dominates — But Only for Action & Sports

North America drives \$4,401M (49.3%) of all global sales with a massive bias toward Action (\$879M) and Sports titles.

02 Japan is an RPG Island

Japan completely rejects Action/Shooter games. RPGs generate \$355.4M there — while Action games generate only a fraction. Same genre, opposite results.

03 The Industry Peaked in 2008 — and Never Recovered

Global sales hit \$704.3M in 2008 then entered terminal decline. Publishers can no longer rely on market growth — they must steal share.

04 Critics ≠ Commercial Success ($R^2 = 0.06$)

Wii Sports sold 82.53M units without a 90+ score. Mega-hits are driven by hardware install base and marketing — not prestige.

05 PS2 is the Undisputed Volume King

PlayStation 2 accounts for \$1,255.6M in software sales — nearly 2× its nearest competitor. Hardware install base is the real multiplier.

06 Nintendo, EA & Activision Control the Market

Publisher revenue follows a Pareto distribution. Mid-tier publishers cannot compete head-to-head in AAA Action — niche genre focus is survival strategy.

Statistical Proof Behind the Insights

Reference: notebooks/04_statistical_analysis.ipynb · `scipy` · `statsmodels` · `sklearn`

OLS REGRESSION

```
statsmodels
```

$$R^2 = 0.060$$

Critic & User scores together explain only 6% of global sales variance. Mega-hits are driven by marketing spend, platform install base & franchise IP — not review scores.

HYPOTHESIS TESTING

```
scipy.stats → t-test
```

$$T = 14.86 \cdot p < 0.001$$

NA and EU markets are statistically proven to be fundamentally different revenue streams. Treating them identically is a costly statistical error.

CORRELATION MATRIX

```
Pearson + Spearman
```

$$\text{User Score} \leftrightarrow \text{Sales: } r = 0.088$$

User sentiment has near-zero linear relationship with sales. Publishers who use early user reviews to forecast commercial success are operating on a false signal.

SCENARIO ANALYSIS

```
OLS regression outputs
```

Critic-pleasing = Diminishing Returns

Simulating disproportionate investment in 'critic-pleasing' mechanics shows severely diminishing ROI. Once a baseline quality (~75+) is reached, only scale & marketing move volume.

Three Views. One Story.

Live: [Video Game Sales Dashboard](#)

VIEW 1 Executive Overview

Audience: CEO / C-Suite

Total sales KPIs · Regional donut · Top 10 games · Historical sales trend (peak: 2008)

VIEW 2 Genre & Platform Deep Dive

Audience: Portfolio Manager

Platform sales ranking · Genre Critic Scores · Genre by Era (area chart) · Genre Sales bar

VIEW 3 Regional & Ratings Analysis

Audience: Marketing Director

Regional by Genre (NA vs JP divide) · Score Gap histogram · Critic vs Sales scatter · Top Publishers treemap



INTERACTIVE FILTERS: Platform · Genre · Era (Decade) · ESRB Rating | Cross-filtering + Drill-down tooltips

From Data to Decisions

Each recommendation directly linked to a statistically proven insight

INSIGHT (from data)	→ RECOMMENDATION	EXPECTED IMPACT
NA/EU buys Action · JP buys RPGs · uniform global campaigns waste ad spend	Allocate 90%+ of Action/Shooter budgets to NA/EU. Reserve RPG spend exclusively for Japan. Kill global campaigns.	Significantly higher ROAS · No global ad waste
Japanese localization for Western Shooters yields <5% of global volume	Cease JP translation, dubbing & local marketing for all FPS/Action titles immediately.	\$1M–\$3M saved per AAA title · <5% drop in global sales
Critic Scores explain only 6% of sales variance ($R^2 = 0.06$)	Replace Metacritic score bonuses with Day-30 User Engagement & Sentiment KPIs for studio evaluation.	Better community retention · Longer, more profitable sales tails
PS2, X360, PS3 dominated because they had the largest install base at peak	Launch mid-tier & AAA titles strictly on platforms at peak adoption. Never risk Year-1 launches on unproven hardware.	Maximum addressable audience · Lower launch risk

The Cost of Inaction

AAA development exceeds \$100M. Every misspent dollar is a dollar that doesn't return.

\$1M–\$3M

Saved per AAA title by cutting JP localization for Shooters

+10–15%

Estimated NA revenue lift by reinvesting localization savings

Only 6%

Of global sales variance explained by critic scores ($R^2 = 0.06$)

<5%

Drop in global volume from removing JP Shooter localization

WHY ACT NOW

- Physical market peaked in 2008 — the window of easy growth is permanently closed.
- AAA budgets routinely exceed \$100M; mistargetted ad spend compounds losses at scale.
- Streaming & digital have erased the margin for error in physical launch strategies.
- Publishers who pivot to data-driven regional allocation first gain permanent competitive advantage.

What We Know — and What's Next

Honest scope boundaries + the roadmap forward

LIMITATIONS

The Digital Blindspot

Data caps at Dec 2016, physical retail only. Steam, PSN & Xbox Live are completely excluded — severely undercounting post-2013 PC and console performance.

Review Imputation Bias

50%+ of review scores were imputed using column medians (Critic: 71.0 · User: 7.50). This artificially centralises the rating distribution.

No Profitability Data

Development budgets and marketing costs are absent. We can conclude gross volume popularity — not net financial ROI.

NEXT STEPS

01

New Data Sources

Procure SteamSpy / AppMagic digital sales + microtransaction/DLC revenue datasets to complete the post-2016 commercial picture.

02

Predictive Modelling

Build a Random Forest or XGBoost classifier to predict probability of a new game pitch crossing 1M sales — based on Genre, Platform, Release Month.

03

Real-Time Dashboard

Connect Tableau to IGDB or Twitch Viewership API for live user sentiment and player counts — replacing static CSV uploads.

Contributions

SecE_G15 · Newton School of Technology · Faculty: Ayushi Vashisht

	Dataset Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
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Ankita Thakur	✓	✓				✓	✓
Sarabjeet Singh	✓				✓		✓
Anuj Upadhyay	✓			✓			✓
Aaditya Yadav	✓		✓		✓		✓